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Math 400

Section 5.5 Practice Problems

Chapter 5 Review Exercises page 422 # 14, 17, 18, 19, 20, 21, 24, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38

1. Evaluate each indefinite integral.

$(a.)\ \int (sec^2 3x)\, dx$

$(b.)\ \int \frac{e^x}{\sqrt{4-e^{2x}}} dx$

$(c.)\ \int \left(x^4 e^{x^5}\right) dx$

$(d.)\ \int \frac{\cos(\ln x)}{x} dx$

(e.) $\int \frac{\arctan^3 x}{x^2+1} dx$

$(f.)\ \int x^2\sqrt{x+1}\, dx$

$(g.)\ \int \tan x \ln(\cos x)\, dx$

$(h.)\ \int \frac{x}{\sqrt{1-x^4}} dx$

$(i.)\ \int (1-x)\sqrt{2x-x^2}\, dx$

$(j.)\ \int (x^2+2x)\sqrt{x+3}\, dx$

$Hint\colon\ (x^2+2x)\sqrt{x+3} = x^2\sqrt{x+3} + 2x\sqrt{x+3}$

$(k.)\ \int \sin x \cos(\cos x)\, dx$

$(l.)\ \int 2^{1+\sinh x}(\cosh x)\, dx$

2. Evaluate each definite integral.

$(a.)\ \int_0^2 x^2(1+2x^3)^3\,dx$

$(b.)\ \int_0^{\pi/4} (1+\tan x)^2 sec^2\,x\,dx$

$(c.)\ \int_0^1 \frac{x^2}{\sqrt{4-3x}}dx$

$(d.)\ \int_{e^2}^{e^3} \frac{lnx}{x}dx$

$(e.)\ \int_0^1 \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}dx$